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In [6]: #EXP 3- CLASSIFICATION WITH DECISION TREES
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import confusion_matrix, accuracy_score
import matplotlib.pyplot as plt
import seaborn as sns

iris = load_iris()
X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

dt_model = DecisionTreeClassifier(criterion='gini', random_state=0)
dt_model.fit(X_train, y_train)

# Step 5: Predict the Test Data
y_pred = dt_model.predict(X_test)

# Step 6: Evaluate the Model
cm = confusion_matrix(y_test, y_pred)
acc = accuracy_score(y_test, y_pred)

print("Confusion Matrix:")
print(cm)
print("\nAccuracy Score:", acc)

plt.figure(figsize=(6,4))
sns.heatmap(
    cm,
    annot=True,
    cmap="Blues",
    xticklabels=iris.target_names,
    yticklabels=iris.target_names
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()

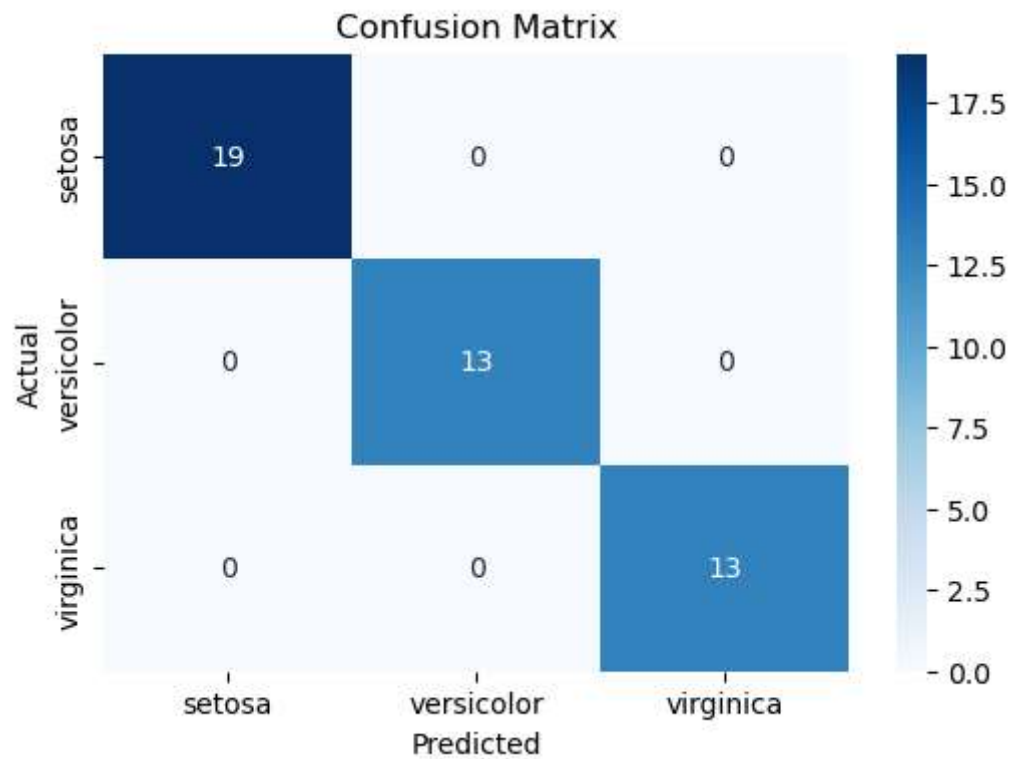
plt.figure(figsize=(12,8))
plot_tree(
    dt_model,
    filled=True,
    feature_names=iris.feature_names,
    class_names=iris.target_names
)
```

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plt.title("Decision Tree Visualization")  
plt.show()
```

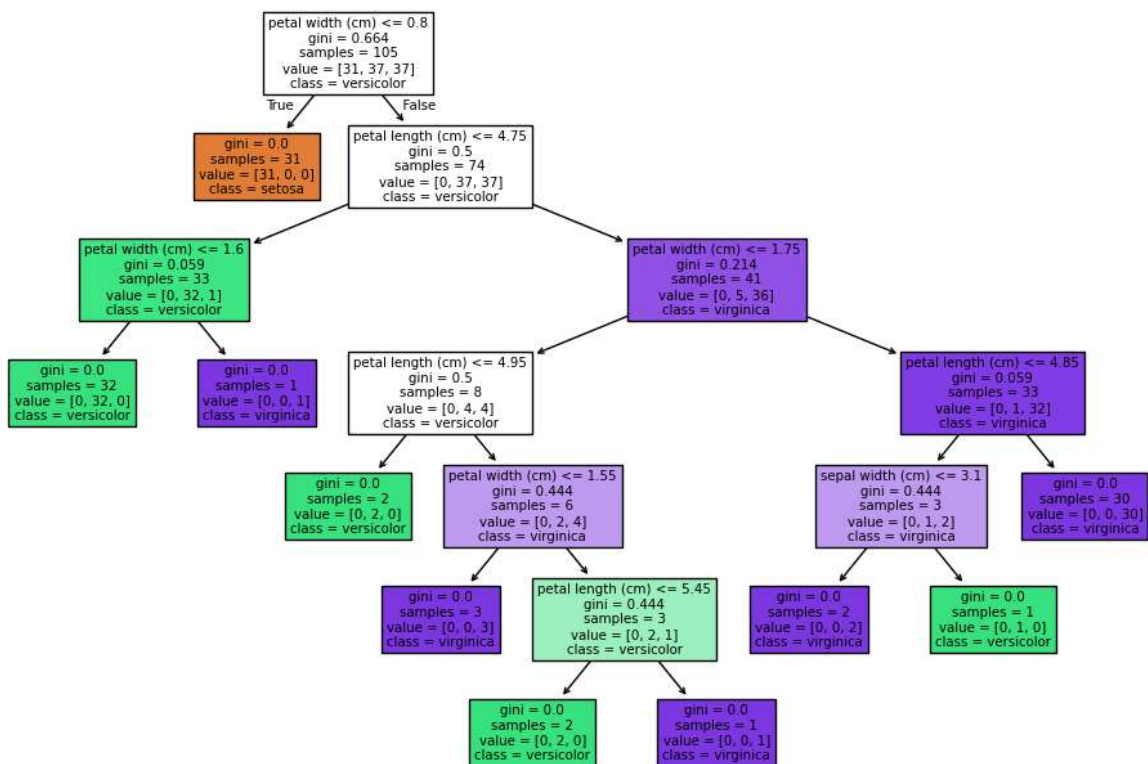
Confusion Matrix:

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[[19  0  0]  
 [ 0 13  0]  
 [ 0  0 13]]
```

Accuracy Score: 1.0



Decision Tree Visualization



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In [7]: #EXP-3 MINI PROJECT
#STUDENT PERFORMANCE PREDICTION (PASS OR FAIL)
#DATASET....
import pandas as pd

data = {
    'Study_Hours':[2,5,1,6,4,3,7,2,5,4,1,6,3,7,2,5,4,1,6,3],
    'Attendance':[60,85,50,90,75,65,95,55,80,70,45,88,68,92,58,82,72,40,87,66],
    'Assignments_Completed':[3,8,2,9,6,4,10,3,7,5,1,8,4,9,2,7,6,1,8,4],
    'Result':[0,1,0,1,1,0,1,0,1,1,0,1,0,1,1,0,1,0,1,0]
}

df = pd.DataFrame(data)
df.to_csv("student_performance.csv", index=False)

print("Dataset created successfully!")

```

Dataset created successfully!

```

In [8]: # Student Performance Prediction Using Decision Tree

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import confusion_matrix, accuracy_score

# Load Dataset

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data = pd.read_csv("student_performance.csv")

# Features and Target
X = data[['Study_Hours', 'Attendance', 'Assignments_Completed']]
y = data['Result']

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Train Decision Tree
model = DecisionTreeClassifier(
    criterion='gini',
    random_state=0,
    max_depth=3
)

model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy Score:", accuracy)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

# Confusion Matrix Visualization
plt.figure(figsize=(6, 5))
sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=['Fail', 'Pass'],
    yticklabels=['Fail', 'Pass']
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()

# Decision Tree Visualization
plt.figure(figsize=(18, 10))

plot_tree(
    model,
    feature_names=X.columns,
    class_names=['Fail', 'Pass'],
```

```
        filled=True,
        rounded=True,
        fontsize=12
    )

plt.title("Student Performance Decision Tree", fontsize=16)
plt.tight_layout()

# Save High Quality Image
plt.savefig(
    "decision_tree.png",
    dpi=300,
    bbox_inches='tight'
)

plt.show()

# New Student Prediction
new_student = pd.DataFrame(
    {
        'Study_Hours': [5],
        'Attendance': [80],
        'Assignments_Completed': [7]
    }
)

prediction = model.predict(new_student)

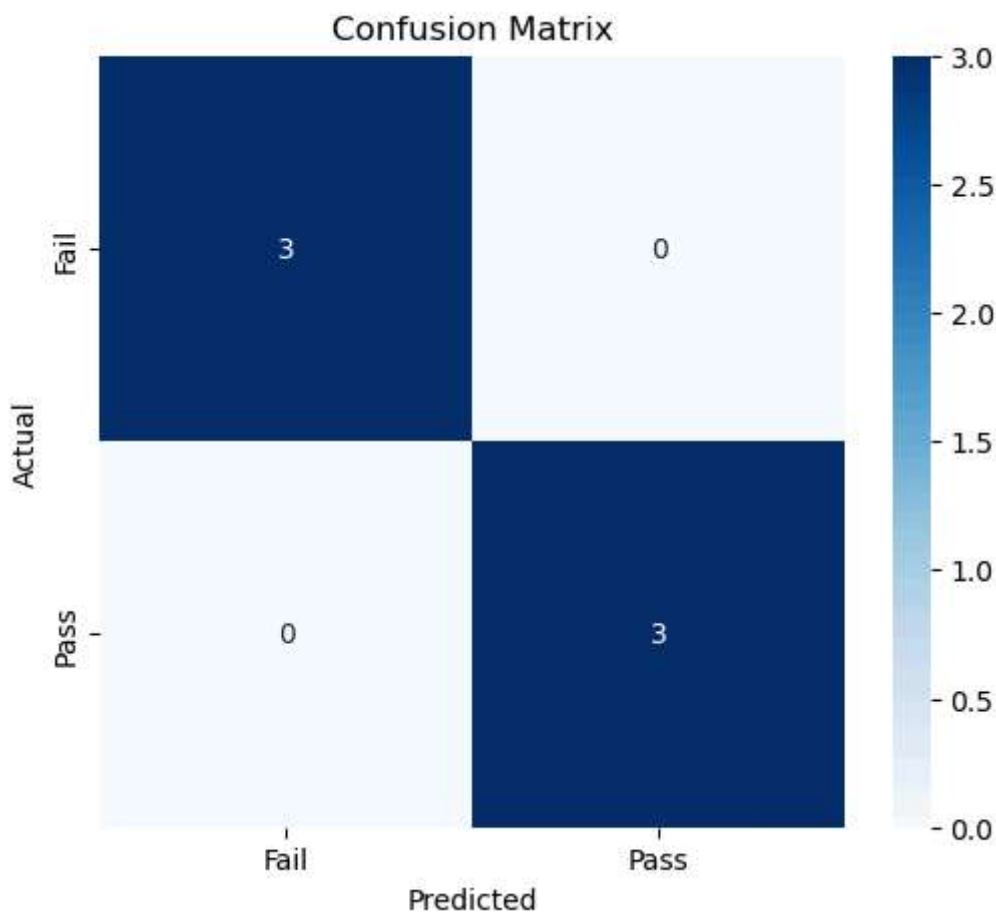
print("\nNew Student Data:")
print(new_student)

if prediction[0] == 1:
    print("\nPrediction: PASS")
else:
    print("\nPrediction: FAIL")
```

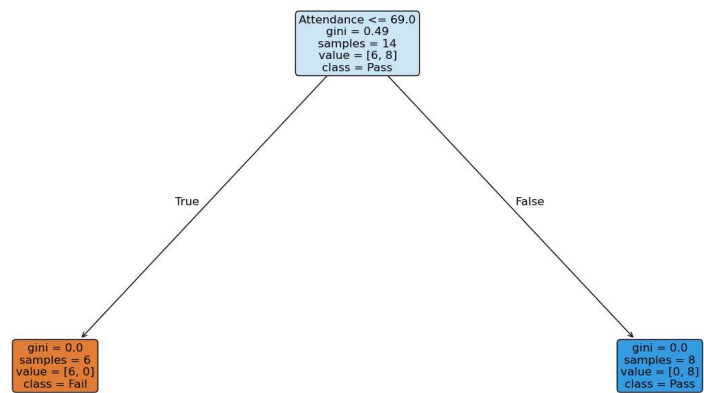
Accuracy Score: 1.0

Confusion Matrix:

```
[[3 0]
 [0 3]]
```



Student Performance Decision Tree



New Student Data:

Study_Hours	Attendance	Assignments_Completed
0	5	80
		7

Prediction: PASS